

Character-Theoretic Diagonalization of the Modulo-420 Channel Asymmetry in Prime Quadruplets

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Abstract

We study the occupancy asymmetry on the six observed prime-quadruplet half-channels $\mathcal{R} = \{11, 101, 191, 221, 311, 401\}$ modulo 420. Three methodological levels structure the analysis. **Level 1 (exact algebra):** the primitive cubic Dirichlet character $\chi_3 \bmod 7$ coincides pointwise with the $k=2$ discrete Fourier mode on $\mathcal{R}$ ($\chi_3|_{\mathcal{R}} = \psi_2$), hence $F_2 = \langle \delta, \chi_3 \rangle$ for the observed bias vector δ ; the bias admits an exact orthogonal decomposition into Dirichlet character components modulo 420. **Level 2 (numerical, conditional on H):** under a multinomial equidistribution hypothesis H , the χ_3 -projection scales as $|\langle \delta_N, \chi_3 \rangle| \sim 1/(6\sqrt{N})$ with z -scores ≈ 1 at $N_{\max} \in \{10^8, 3 \cdot 10^8, 10^9\}$ —pure fluctuation, no persistent Chebyshev-type drift. **Level 3 (HL corroboration):** a Hardy–Littlewood singular-series analysis yields *identical* coefficients on all six channels in $\mathcal{R}$, predicting asymptotic equidistribution $1/6$; the observed asymmetry is statistically consistent with finite-sample fluctuations ($\chi^2 \approx 4.34$, $p \approx 0.50$). Analytic proof of conditional equidistribution remains open.

1 Three-Level Framework

The present work separates three epistemically distinct layers. Only Level 1 consists of unconditional exact statements; Level 2 is probabilistic and conditional on Hypothesis H ; Level 3 remains open.

1.1 Level 1: Exact Algebraic Statements

These results hold for the *fixed finite channel set* $\mathcal{R}$ and the *observed sample* bias δ (or δ_N at any finite count N). They do not assert asymptotic bias in N .

1. **Pointwise identity.** $\chi_3|_{\mathcal{R}} = \psi_2$ as vectors in $\mathbb{C}^6$, where $\psi_k(n) := e^{2\pi i kn/6}$ in the fixed order $(11, 101, 191, 221, 311, 401)$.
2. **DFT–character coincidence.** For the observed (centered) bias vector δ , $F_2 = \langle \delta, \chi_3 \rangle$, with F_k the k -th DFT coefficient on the six-point set.
3. **Orthogonal decomposition.** δ lies in the Dirichlet character subspace mod 420: numerically, $\delta = P_{\chi_3}\delta + P_{\overline{\chi_3}}\delta + P_{\chi_4}\delta + P_{\chi_4\chi_3}\delta + \varepsilon$ with $\|\varepsilon\|^2 < 10^{-36}$ for the sample at $N = 4767$.

1.2 Level 2: Numerically Supported Under Hypothesis H

Under the multinomial null model H (Section 5), the χ_3 -coefficient is a fluctuation of typical size $f(N) = 1/(6\sqrt{N})$:

$$|\langle \delta_N, \chi_3 \rangle| \sim \frac{1}{6\sqrt{N}}, \quad z \approx 1,$$

at $N_{\max} \in \{10^8, 3 \cdot 10^8, 10^9\}$. The pair bias $B_{101,11}/\sqrt{N}$ shows no stable positive drift. The data are compatible with multinomial fluctuations around the uniform distribution, not with a growing structural bias.

1.3 Level 3: HL Corroboration and Open Equidistribution

Section 6 computes the Hardy–Littlewood singular series for the gap constellation $(0, 2, 6, 8)$ and shows that all six channels in $\mathcal{R}$ carry identical local factors mod $2, 3, 4, 5, 7$. The HL heuristic therefore predicts asymptotic equidistribution $P(p \bmod 420 = r \mid Q(p) \text{ prime}) \rightarrow 1/6$. The observed occupancy vector at $N = 4767$ is statistically consistent with this prediction ($\chi^2 \approx 4.34$, $p \approx 0.50$). What remains *unproved* is an analytic equidistribution theorem for $p \bmod 420$ conditional on $Q(p)$ being prime, including control of arithmetic correlations between successive events.

2 Setup and Notation

Let $Q(p) = (p, p+2, p+6, p+8)$ denote a prime quadruplet with gap signature $(2, 4, 2)$. The six observed half-channels modulo 420 are

$$\mathcal{R} = \{11, 101, 191, 221, 311, 401\}.$$

For N counted quadruplet events with channel label $r_k \in \mathcal{R}$, write $n_r(N)$ for occupancy counts and $K = \sum_r n_r(N) = N$.

The *centered channel bias* (probability form) is

$$\delta_N(r) := \frac{n_r(N)}{N} - \frac{1}{6}, \quad r \in \mathcal{R}.$$

For the reference sample $N = 4767$ we write $\delta := \delta_N$. The inner product on $\mathbb{C}^{\mathcal{R}}$ is

$$\langle f, g \rangle := \frac{1}{6} \sum_{r \in \mathcal{R}} f_r \overline{g_r}.$$

Dirichlet characters mod $420 = 4 \cdot 3 \cdot 5 \cdot 7$ are written as product characters $\chi = \chi_4 \cdot \chi_5 \cdot \chi_7^{(q)} \cdot \chi_7^{(3)}$, with χ_4 the primitive quadratic character mod 4, $\chi_5 = (\cdot/5)$, $\chi_7^{(q)} = (\cdot/7)$, and $\chi_7^{(3)} = \chi_3$ the primitive *cubic* character mod 7 ($\chi_3(\zeta) = \zeta$ for $\zeta \in \mathbb{F}_7^\times$, $\omega = e^{2\pi i/3}$, $\chi_3(4) = 1$, $\chi_3(3) = \omega$, $\chi_3(2) = \omega^2$).

3 Character-Theoretic Diagonalization

We state only what is proved or numerically verified for the *observed finite sample*; coefficients do not grow with N in the algebraic layer.

Lemma 1 (Mod-7 cycle on $\mathcal{R}$). *In the fixed channel order $(11, 101, 191, 221, 311, 401)$,*

$$r_n \bmod 7 = (4, 3, 2, 4, 3, 2).$$

Proof. Direct computation: $11 \equiv 4, 101 \equiv 3, 191 \equiv 2, 221 \equiv 4, 311 \equiv 3, 401 \equiv 2 \pmod{7}$. □

Lemma 2 (Pointwise identity $\chi_3 = \psi_2$ on $\mathcal{R}$). *With $\omega = e^{2\pi i/3}$,*

$$\chi_3(r_n) = \psi_2(n) \quad \text{for all } n = 0, \dots, 5.$$

Hence $\chi_3|_{\mathcal{R}}$ and ψ_2 are identical vectors in $\mathbb{C}^6$ (not merely collinear).

Proof. By Lemma 1, $(\chi_3(r_n))_{n=0}^5 = (1, \omega, \omega^2, 1, \omega, \omega^2)$. Since $\psi_2(n) = e^{2\pi i \cdot 2n/6} = e^{2\pi i n/3}$, the sequences agree. $\square$

Theorem 3 (Character-Theoretic Diagonalization). (For the observed finite sample on $\mathcal{R}$.) *Let $(\delta_n)_{n=0}^5$ be δ in the fixed channel order and*

$$F_k := \frac{1}{6} \sum_{n=0}^5 \delta_n e^{-2\pi i k n/6} = \langle \delta, \psi_k \rangle$$

the k -th DFT coefficient. Then:

1. **Exact coincidence.** $F_2 = \langle \delta, \chi_3 \rangle$. Since $\delta \in \mathbb{R}^6$, also $F_4 = \overline{F_2}$.
2. **Energy share.** The projection onto $\text{span}\{\chi_3, \overline{\chi_3}\}$ accounts for $|F_2|^2 + |F_4|^2 = |\langle \delta, \chi_3 \rangle|^2 + |\langle \delta, \overline{\chi_3} \rangle|^2 = 0.5971 \cdot \text{Var}(\delta)$ in the sample at $N = 4767$.
3. **Complete character decomposition.** With orthonormal character directions $\{\hat{e}_\chi\}$ on $\mathcal{R}$ (Gram-Schmidt on character values),

$$\delta = \sum_{\chi} c_{\chi} \hat{e}_{\chi}, \quad c_{\chi} = \langle \delta, \hat{e}_{\chi} \rangle, \quad \text{Var}(\delta) = \sum_{\chi} |c_{\chi}|^2.$$

Numerically,

$$\delta = P_{\chi_3} \delta + P_{\overline{\chi_3}} \delta + P_{\chi_4} \delta + P_{\chi_4 \chi_3} \delta + \varepsilon, \quad \|\varepsilon\|^2 < 10^{-36}.$$

The residual after removing $\chi_3 + \overline{\chi_3}$ is carried primarily by χ_4 (10.67% of total variance) and $\chi_4 \chi_3$ (28.75%); χ_5 vanishes since $r \equiv 1 \pmod{5}$ on all channels.

Proof. Part (1). Lemma 2 gives $\psi_2 = \chi_3|_{\mathcal{R}}$, hence $F_2 = \langle \delta, \psi_2 \rangle = \langle \delta, \chi_3 \rangle$. Reality of δ and $\psi_4 = \overline{\psi_2}$ yield $F_4 = \overline{F_2}$.

Part (2). Parseval on the six-point set; modes $k = 2, 4$ span the same two-dimensional cubic subspace as $\{\chi_3, \overline{\chi_3}\}$.

Part (3). All $r \in \mathcal{R}$ are odd and $\not\equiv 0 \pmod{5, 7}$, so all characters are well-defined. On $\mathcal{R}$, $r \equiv 1 \pmod{5}$ and $r \equiv 2 \pmod{3}$, hence χ_5 and the quadratic character mod 3 are constant and contribute no variance. Gram-Schmidt reconstruction gives $\text{Var}(\varepsilon) = 1.31 \times 10^{-36}$. On the $\chi_3 = 1$ subclass $\{11, 221\}$, only χ_4 separates the two channels ($\chi_4(11) = -1$, $\chi_4(221) = +1$). $\square$

Remark 1 (Scope of the theorem). *Theorem 3 is a diagonalization of fluctuations in character coordinates for the observed finite sample. It identifies χ_3 as the dominant Fourier direction on $\mathcal{R}$ by algebraic coincidence, not by asymptotic equidistribution. It does not claim that $|\langle \delta_N, \chi_3 \rangle|$ grows with N , nor that a Chebyshev-type bias persists.*

4 Numerical Results

4.1 Character decomposition at $N_{\max} = 10^8$ ($K = 4767$)

4.2 Scaling across $10^8, 3 \cdot 10^8, 10^9$

The product $|\langle \delta, \chi_3 \rangle| \sqrt{K} \approx 0.18 \pm 0.01$ is stable across three decades; all z -scores lie near 1 (Rayleigh median ≈ 1.18 under H). The ratio $B/\sqrt{K}$ decreases monotonically ($0.956 \rightarrow 0.487 \rightarrow 0.457$) without stabilizing at a positive constant; the terminal null p -value is $\approx 43\%$.

Table 1: Orthonormal character projections of δ on $\mathcal{R}$; variance shares in percent ($\text{Var}(\delta) = 2.53 \times 10^{-5}$).

Character χ	$ c_\chi $	$\arg(c_\chi)$	Variance (%)
1 (trivial)	0.00000000	—	0.0000
χ_3	0.00274918	-106.63°	29.8566
$\overline{\chi_3}$	0.00274918	$+106.63^\circ$	29.8566
χ_4	0.00164324	0.00°	10.6669
$\chi_4 \chi_7^{(q)}$	0.00046972	180.00°	0.8716
$\chi_4 \chi_3$	0.00269768	-30.00°	28.7484
Sum	—	—	100.0000

Table 2: Primitive character values on $\mathcal{R}$.

r	$r \bmod 4$	$r \bmod 5$	$r \bmod 7$	χ_4	χ_5	χ_3
11	3	1	4	-1	+1	+1
101	1	1	3	+1	+1	ω
191	3	1	2	-1	+1	ω^2
221	1	1	4	+1	+1	+1
311	3	1	3	-1	+1	ω
401	1	1	2	+1	+1	ω^2

5 Proof Attempt: Hypothesis H and $f(N) = 1/(6\sqrt{N})$

Hypothesis 1 (H —Multinomial equidistribution on $\mathcal{R}$). *For N quadruplet events, channel occupancy follows*

$$(n_{11}, n_{101}, \dots, n_{401}) \sim \text{Mult}(N; \frac{1}{6}, \dots, \frac{1}{6}).$$

Equivalently, channel indices $(I_k)_{k=1}^N$ are i.i.d. uniform on $\{1, \dots, 6\}$.

Remark 2. H is the asymptotic consequence of Hardy–Littlewood equidistribution if all six channels in $\mathcal{R}$ carry equal singular-series factors. H does not postulate Chebyshev drift, rank-one dynamics, or causal EABC interpretation.

Proposition 4 (CLT scaling under H). *Under Hypothesis 1, as $N \rightarrow \infty$:*

1. $\mathbb{E}[\langle \delta_N, \chi_3 \rangle] = 0$;
2. $\text{Var}(\langle \delta_N, \chi_3 \rangle) = 1/(36N)$;
3. $\sqrt{N} \langle \delta_N, \chi_3 \rangle \xrightarrow{d} \mathcal{N}_{\mathbb{C}}(0, 1/36)$;
4. $|\langle \delta_N, \chi_3 \rangle| = O_p(1/\sqrt{N})$.

Proof attempt (conditional on H). Since $\sum_{r \in \mathcal{R}} \chi_3(r) = 0$, the mean cancels:

$$\langle \delta_N, \chi_3 \rangle = \frac{1}{6N} \sum_{r \in \mathcal{R}} n_r \overline{\chi_3(r)} = \frac{S_N}{6N}, \quad S_N := \sum_{k=1}^N \overline{\chi_3(r_{I_k})}.$$

Under H , $Y_k := \overline{\chi_3(r_{I_k})}$ are i.i.d. with $\mathbb{E}[Y_k] = 0$ and $\mathbb{E}[|Y_k|^2] = 1$, hence $\text{Var}(S_N) = N$ and $\text{Var}(S_N/(6N)) = 1/(36N)$. The multidimensional CLT gives $\sqrt{N} \langle \delta_N, \chi_3 \rangle \xrightarrow{d} \mathcal{N}_{\mathbb{C}}(0, 1/36)$. $\square$

Table 3: χ_3 -projection vs. theory $f(K) = 1/(6\sqrt{K})$ under H (data from `eabc_analytic_proof_check.py`; multinomial null: 10 000 trials).

$N_{\max}$	K	$ \langle \delta, \chi_3 \rangle $	$f(K)$	z	$ \langle \delta, \chi_3 \rangle \sqrt{K}$	$p(\geq)$
10^8	4767	0.002749	0.002414	1.14	0.190	26.8%
$3 \cdot 10^8$	10972	0.001568	0.001591	0.99	0.164	37.7%
10^9	28387	0.001043	0.000989	1.05	0.176	33.1%

Table 4: Pair bias $B_{101,11} = N_{101} - N_{11}$ (data from `eabc_scaling_test.py`).

$N_{\max}$	K	$B_{101,11}$	$B/\sqrt{K}$
10^8	4767	+66	+0.956
$3 \cdot 10^8$	10972	+51	+0.487
10^9	28387	+77	+0.457

Corollary 5 (Explicit scaling function). *Under H , the natural scale is*

$$\boxed{f(N) = \frac{1}{6\sqrt{N}}}, \quad \langle \delta_N, \chi_3 \rangle \sim \mathcal{N}_{\mathbb{C}}\left(0, \frac{1}{36N}\right).$$

The z -score $|\langle \delta_N, \chi_3 \rangle|/f(N)$ is asymptotically Rayleigh(1)-distributed (median ≈ 1.177). Expected magnitude: $\mathbb{E}[|\langle \delta_N, \chi_3 \rangle|] \approx \sqrt{\pi/(72N)} \approx 0.886 f(N)$.

Proof gap. The CLT argument is complete *conditional on H* . Section 6 supplies arithmetic evidence that H is the natural HL consequence on $\mathcal{R}$; an analytic proof of conditional equidistribution (Hypothesis H6) remains open.

6 Hardy–Littlewood Equidistribution Analysis

Let $Q = (0, 2, 6, 8)$ denote the gap constellation for quadruplets $(p, p+2, p+6, p+8)$ and

$$\pi_Q(x) := \#\{p \leq x : p, p+2, p+6, p+8 \text{ prime}\}.$$

The Hardy–Littlewood conjecture [4] (Bateman–Horn form) predicts

$$\pi_Q(x) \sim C_Q \int_2^x \frac{dt}{\log^4 t}, \quad C_Q = 2 \cdot \mathfrak{S}(Q),$$

where $\mathfrak{S}(Q) = \prod_p w_p(Q)$ is the *singular series*. For prime p , let ν_p be the number of forbidden start classes mod p (classes n with $n+h \equiv 0$ for some $h \in Q$). Then

$$w_p(Q) = 1 - \frac{\nu_p}{(p-1)^4}. \tag{1}$$

A partial product to $p \leq 2 \cdot 10^5$ gives $\mathfrak{S}(Q) \approx 0.858$.

Conditional distribution mod q . For modulus q , let ω_q be the number of admissible start classes ($n \bmod q$ with all four positions $\not\equiv 0$). Under HL equidistribution on admissible classes, $p \equiv r \bmod q$ carries relative weight $1/\omega_q$ if admissible, else 0. The total Hardy–Littlewood weight of channel $r \in \mathcal{R}$ is

$$W_{\text{HL}}(r) = \prod_{q \in \{2,3,4,5,7\}} \frac{1}{\omega_q} \cdot \prod_{p \in \{3,5,7\}} w_p(Q), \tag{2}$$

provided r is admissible at every factor.

Lemma 6 ($\mathcal{R}$ is complete mod 420). $\omega_{420} = 6$, and the only admissible residue classes are exactly $\mathcal{R} = \{11, 101, 191, 221, 311, 401\}$.

Proof. By the Chinese remainder theorem, $\omega_{420} = \omega_4 \cdot \omega_3 \cdot \omega_5 \cdot \omega_7 = 2 \cdot 1 \cdot 1 \cdot 3 = 6$. Direct enumeration confirms $\mathcal{R}$ as the six classes (`eabc_hl_coefficient_hypotheses.py`). $\square$

Theorem 7 (Identical HL coefficients on $\mathcal{R}$). For all $r \in \mathcal{R}$, $W_{\text{HL}}(r) = W_0$ for the same constant W_0 . In particular, the HL heuristic predicts $P(p \bmod 420 = r \mid Q(p) \text{ prime}) \rightarrow 1/6$.

Proof. Every $r \in \mathcal{R}$ satisfies: $r \equiv 1 \bmod 2$ ($\omega_2 = 1$); $r \equiv 2 \bmod 3$ ($\omega_3 = 1$); $r \equiv 1$ or $3 \bmod 4$, three channels each ($\omega_4 = 2$); $r \equiv 1 \bmod 5$ ($\omega_5 = 1$); $r \equiv 2, 3$ or $4 \bmod 7$, two channels each ($\omega_7 = 3$). Each factor in (2) is therefore identical across $\mathcal{R}$. $\square$

6.1 Local factors and goodness of fit

Table 5 lists local congruence structure, observed occupancies at $N = 4767$ ($p \leq 10^8$), and identical HL weights. Small-prime singular factors are $w_3 = 7/8$, $w_5 = 63/64$, $w_7 \approx 0.997$.

Table 5: Local HL structure and channel weights ($N = 4767$, $p \leq 10^8$; data from `eabc_hl_coefficient_hypotheses.py`).

r	n_r	$\hat{p}_r$	δ_r	$r \bmod 4$	$r \bmod 5$	$r \bmod 7$	$W_{\text{HL}}(r)$	$W_{\text{HL}}/\sum$
11	765	0.1605	-0.00619	3	1	4	3.47×10^{-4}	1/6
101	831	0.1743	+0.00766	1	1	3	3.47×10^{-4}	1/6
191	786	0.1649	-0.00178	3	1	2	3.47×10^{-4}	1/6
221	809	0.1697	+0.00304	1	1	4	3.47×10^{-4}	1/6
311	809	0.1697	+0.00304	3	1	3	3.47×10^{-4}	1/6
401	767	0.1609	-0.00577	1	1	2	3.47×10^{-4}	1/6
Local moduli: $\omega_2=1$, $\omega_3=1$, $\omega_4=2$, $\omega_5=1$, $\omega_7=3$.								

Table 6: HL prediction vs. observation ($N = 4767$).

Quantity	Value
χ^2 (HL uniform 1/6)	4.344
p -value (5 df)	0.501
$\text{KL}(\hat{\mathbf{p}} \parallel \mathbf{p}_{\text{HL}})$	4.55×10^{-4}
$\max_i \delta_i $	0.77%
$ \langle \delta, \chi_3 \rangle $	0.00275
$ \langle \delta, \chi_3 \rangle /(1/(6\sqrt{N}))$	≈ 1.1

Since all HL weights coincide, $\mathbf{p}_{\text{HL}}$ and the uniform prediction 1/6 are identical; a channel-specific HL correction does not improve the fit.

6.2 Hypotheses H1–H6

We summarize six testable hypotheses (full analysis in `eabc_hl_hypotheses.tex`):

H1 (identical HL coefficients). All six channels carry the same singular-series weight (Theorem 7); deviations $\leq 0.77\%$. *Supported.*

H2 (mod-7 HL densities). Different $p \bmod 7$ classes induce distinct HL factors and explain the χ_3 component. *Refuted:* $\omega_7 = 3$ yields equal factors 1/3 per class; $\chi_3|_{\mathcal{R}} = \psi_2$ is algebraic, not density-theoretic.

- H3 (mod-4/mod-5 HL densities).** Alternating $r \bmod 4$ or constant $r \bmod 5$ induce distinct HL weights feeding the χ_4 projection. *Refuted:* $\omega_4 = 2$, $\omega_5 = 1$ give identical factors on $\mathcal{R}$.
- H4 (inter-quadruplet correlations).** Successive quadruplets are not i.i.d.; transition dynamics modify the channel distribution. *Open, but unlikely* as the main cause: scaling to 10^9 shows no stable drift beyond $O(1/\sqrt{N})$.
- H5 (finite-sample fluctuations).** The observed asymmetry at $N = 4767$ is pure finite-sample noise around HL equidistribution. *Supported:* $\chi^2 \approx 4.34$, $p \approx 0.50$, consistent with Hypothesis 1.
- H6 (unproved analytic equidistribution).** Even with identical coefficients, arithmetic correlations may distort the conditional distribution $p \bmod 420 \mid Q(p)$. *Mathematically open*, numerically unremarkable to 10^9 .

7 Discrete Holonomy Interpretation

This section belongs to **Level 2** of the framework (Section 1): a numerically verified *structural parallel* between the local C_3 -phase pattern induced by χ_3 on $\mathcal{R}$ and a discrete edge-phase model on the six-channel ring. It does not extend the unconditional algebra of Level 1, nor does it assert any Level 3 physical mechanism.

7.1 Discrete connection on the six-channel ring

Fix the channel order $(r_n)_{n=0}^5$ from Section 2. Define vertex phases

$$\theta_n := \arg(\chi_3(r_n)) \in \{0, \frac{2\pi}{3}, -\frac{2\pi}{3}\},$$

with $\arg$ mapped to the principal value in $(-\pi, \pi]$ after rounding to the C_3 phase set (as implemented in `eabc_chi3_holonomy_test.py`). The *discrete connection* along directed edges $n \rightarrow n+1$ (indices mod 6) is

$$\Delta\theta_n := \text{wrap}(\theta_{n+1} - \theta_n), \quad \text{wrap}(\alpha) \in (-\pi, \pi].$$

Global holonomy around the closed ring is

$$H_\chi := \exp\left(i \sum_{n=0}^5 \Delta\theta_n\right), \quad \gamma_\chi := \arg(H_\chi), \quad \text{Flux}_\chi := \frac{\gamma_\chi}{2\pi}.$$

The same construction applies to the $k=2$ Fourier mode $\psi_2(n) = e^{-2\pi i n/3}$ with phases $\phi_n := \arg(\psi_2(n))$ and edge increments $\Delta\phi_n := \text{wrap}(\phi_{n+1} - \phi_n)$.

7.2 Numerical results

Direct computation (`eabc_chi3_holonomy_test.py`; output in `eabc_chi3_holonomy_report.txt`) yields:

- **Local edge phases (χ_3).** $\Delta\theta_n = +2\pi/3$ on all six edges—a constant C_3 -typical step. The first half-cycle ($n = 0, 1, 2$) matches the second ($n = 3, 4, 5$).
- **Local edge phases ($k=2$).** $\Delta\phi_n = -2\pi/3$ on all six edges, the connection-level structural parallel of Lemma 2.

- **Global holonomy.** $\gamma_\chi \approx 0$, $\text{Flux}_\chi \approx 0$ (numerically $\lesssim 10^{-15}$); likewise $\gamma_{\psi_2} \approx 0$, $\text{Flux}_{\psi_2} \approx 0$. The wrapped edge increments sum to an integer multiple of 2π , so the closed six-ring carries trivial total phase holonomy—as expected for a defect-free cycle with no enclosed flux in this discrete model.
- **Null test (χ_5).** Since $\chi_5(r) = +1$ for all $r \in \mathcal{R}$, $\theta_n \equiv 0$ and all $\Delta\theta_n = 0$; holonomy is identically trivial.

These findings confirm a *coordinate-theoretic analogy*: the mod-7 cubic character induces a uniform local phase-transport pattern on the channel ring, while global holonomy cancels on the closed cycle. They mirror the exact identity $\chi_3|_{\mathcal{R}} = \psi_2$ (Level 1) in a graph-theoretic language of edge phases.

Epistemological disclaimer. This establishes a coordinate-theoretic analogy, not a claim about quantized transport or persistent topological bias in prime quadruplets. We do *not* assert quantum Hall plateaus, topological protection in the sense of the von Klitzing effect, an Aharonov–Bohm solenoid experiment, or non-abelian gauge fields realized in nature. The discrete edge-phase model is a structural parallel useful for organizing χ_3 -data on $\mathcal{R}$; no physical mechanism coupling holonomy to quadruplet occupancy has been identified or tested.

8 Conclusion

The modulo-420 channel asymmetry of prime quadruplets admits an exact character-theoretic diagonalization. The dominant Fourier component coincides with the restriction of the cubic Dirichlet character modulo 7, while the remaining structure is accounted for by quadratic character components. However, scaling experiments up to 10^9 show no evidence for a persistent Chebyshev-type drift. The observed amplitudes remain compatible with multinomial fluctuations around the uniform distribution. Consequently, the present data support a character-space decomposition of the fluctuations rather than a genuine asymptotic bias.

A detailed Hardy–Littlewood analysis of the singular series shows identical coefficients for all six modulo-420 channels. Asymptotically, the Hardy–Littlewood heuristic therefore predicts strict equidistribution of prime quadruplets across the channels. The empirically observed asymmetry vector is statistically consistent with pure finite-sample fluctuations ($\chi^2 \approx 4.34$, $p \approx 0.50$). The previously established mathematical dominance of the cubic character χ_3 is consequently not density-theoretic but purely algebraically enforced by the reduction of the underlying residue-class space.

The remaining open problem is an analytic equidistribution theorem for prime quadruplets on the six admissible modulo-420 channels.

Section 7 adds a Level 2 discrete holonomy interpretation: the local C_3 edge-phase pattern induced by χ_3 is numerically consistent with a trivial global holonomy on the closed six-ring, furnishing a coordinate-theoretic analogy only—not evidence for quantized transport or persistent topological bias.

Precision. The diagonalization refers to the *observed finite sample* on $\mathcal{R}$: χ_3 supplies a fixed character coordinate system, not a mechanism whose coefficients grow with N . At $N_{\max} = 10^9$ ($K = 28387$), $|\langle \delta, \chi_3 \rangle|$ scales as $1/\sqrt{K}$ with $z \approx 1$, consistent with $f(N) = 1/(6\sqrt{N})$ under H and inconsistent with a persistent structural bias.

9 Epistemological Scope

The analysis follows a three-stage epistemic chain: *phenomenon* (observed channel asymmetry) $\rightarrow$ *diagonalization* (exact character decomposition, Level 1) $\rightarrow$ *discrete holonomy analogy* (numerical

edge-phase model, Section 7, Level 2) $\rightarrow$ *HL corroboration* (singular-series weights, Level 3). Algebra and asymptotics are deliberately decoupled: Level 1 statements hold unconditionally on the fixed finite set $\mathcal{R}$, while Level 2 and Level 3 address the *magnitudes* of coefficients under probabilistic or arithmetic hypotheses.

- **Confirmed (algebraic diagonalization).** The EABC channel labels organize a finite residue set on which $\chi_3|_{\mathcal{R}} = \psi_2$ holds exactly. Decomposing δ in Dirichlet characters is a proved descriptive tool for the observed fluctuations. The χ_3 dominance is enforced by the residue-class geometry, not by unequal HL densities (H2, H3 refuted).
- **Confirmed (discrete holonomy analogy, Level 2).** On the six-channel ring, χ_3 induces constant local edge phases $\Delta\theta_n = +2\pi/3$ with trivial global holonomy ($\gamma_\chi \approx 0$, $\text{Flux}_\chi \approx 0$); the χ_5 null test is trivial. This is a structural parallel between character phases and a discrete edge-phase model—not a claim about topological protection, quantum Hall plateaus, or physical gauge flux (Section 7).
- **Not confirmed (new asymptotic bias).** Neither $B_{101,11}/\sqrt{N}$ nor $|\langle\delta_N, \chi_3\rangle|$ exhibits growth beyond $O_p(1/\sqrt{N})$. A channel-specific Chebyshev bias is not supported by data to 10^9 .
- **Corroborated (HL singular-series structure).** Theorem 7 shows identical HL coefficients on all six channels; the observed occupancy vector is consistent with uniform $1/6$ ($\chi^2 \approx 4.34$, $p \approx 0.50$, $\text{KL} \approx 4.55 \times 10^{-4}$). This arithmetic input justifies Hypothesis *H* as the natural asymptotic model, without claiming a completed proof.
- **Open (analytic equidistribution, H6).** Proving conditional equidistribution of $p \bmod 420$ given $Q(p)$ prime, and ruling out correlations beyond $O(1/\sqrt{N})$, remains a separate analytic problem.

Diagonalization answers *in which directions* the observed bias decomposes; HL analysis answers *whether unequal arithmetic densities* could account for those magnitudes (they cannot on $\mathcal{R}$); an analytic equidistribution theorem would close the remaining proof gap.

References

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- [4] G. H. Hardy and J. E. Littlewood, *Some problems of ‘Partitio numerorum’ III: On the expression of a number as a sum of primes*, Acta Math. **44** (1923), 1–70.
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A Computational Reproducibility

The numerical claims are reproducible from the following scripts in the same directory:

- `eabc_spectral_decomposition.py` — character decomposition of δ at $K = 4767$; verifies $\chi_3|_{\mathcal{R}} = \psi_2$ and Tables 1–2.
- `eabc_scaling_test.py` — pair bias $B_{101,11}$ and $B/\sqrt{K}$ across $10^8, 3 \cdot 10^8, 10^9$ (Table 4).
- `eabc_analytic_proof_check.py` — χ_3 -projection scaling and multinomial null p -values (Table 3).
- `eabc_chi3_mod7_projection.py` — mod-7 cubic projection diagnostics.
- `eabc_dirichlet_mod420_projection.py` — full mod-420 Dirichlet projection machinery.
- `eabc_cumulative_bias.py` — cumulative bias trajectories.
- `eabc_hl_coefficient_hypotheses.py` — Hardy–Littlewood singular-series computation, local factors mod 2, 3, 4, 5, 7, goodness-of-fit statistics (Tables 5–6; output in `eabc_hl_coefficient_hypotheses_report.txt`).
- `eabc_hl_hypotheses.tex` — standalone HL hypothesis analysis (H1–H6) in German.
- `eabc_chi3_holonomy_test.py` — discrete edge-phase holonomy on the six-channel ring for χ_3 , $k=2$ Fourier mode, and χ_5 null test (Section 7; output in `eabc_chi3_holonomy_report.txt`).